

As3\Josephson Junction.py

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def josephson_simulation(V_func, t_end=10, dt=0.01, phi0=np.pi/2, Ic=1):
5     times = np.arange(0, t_end, dt)
6     phi = phi0
7     I_values = []
8
9     for t in times:
10         V = V_func(t)
11         dphi_dt = V
12         phi += dphi_dt * dt
13         I = Ic * np.sin(phi)
14         I_values.append(I)
15
16     return times, I_values
17
18 # (i) Constant voltage  $V = 5$ 
19 V_constant = lambda t: 5
20 t1, I1 = josephson_simulation(V_constant)
21 plt.plot(t1, I1, '.-', c='green', label='V = 5')
22 #plt.title("Josephson Current with V = 5")
23 plt.xlabel("$Time$")
24 plt.ylabel("$I(t)$")
25 plt.grid(True)
26
27 # (ii) Sinusoidal voltage  $V(t) = 5\cos(t)$ 
28 V_sinusoidal = lambda t: 5 * np.cos(t)
29 t2, I2 = josephson_simulation(V_sinusoidal)
30 plt.plot(t2, I2, '.-', c='red', label='V(t) = 5cos(t)')
31 #plt.title("Josephson Current with V(t) = 5cos(t)")
32 plt.xlabel("$Time$")
33 plt.ylabel("$I(t)$")
34 plt.legend(loc='best')
35
36 plt.axvline(0, c='k')
37 plt.axhline(0, c='k', lw='1.2')
38 plt.grid(True)
39
40 plt.show()
41
```